XIN CAI

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Education

The Chinese University of Hong Kong, Hong Kong, China 2023 – Present
Ph.D. in Information Engineering, expected June 2027 **Supervisor: Prof. Tianfan Xue**

University of Chinese Academy of Sciences (UCAS), Beijing, China 2020 – 2023
M.Eng. in Applied Computer Technology **Supervisor: Prof. Shiguang Shan**

University of Chinese Academy of Sciences (UCAS), Beijing, China 2016 – 2020
B.S. in Computer Science and Technology **Cumulative GPA: 3.90/4.00** **Rank: 2/69**

Research Interests

Generative Models: Unified multimodal models; streaming & long video generation; controllable editing
Vision Representation: Generation-friendly representation for unified models; image/video tokenization
Computational Imaging: Image reconstruction; inverse problems with generative models
3D Computer Vision: 3D reconstruction; rendering and generation

Research Experience

Research Intern **Tencent Hunyuan Multimodal Pretrain Team** Mar. 2026 – Present

- **Led a from-scratch, full-stack pipeline for a mid-scale MoE-DiT text-to-image model** (data construction, MoE architecture, pre-training, evaluation), targeting SOTA open-source level.
- **Primary contributor of the Hunyuan Image 4.0 VAE.** Tencent 2026 Project UP (Qingyun).

Research Scientist Intern **Adobe Research**, San Jose, CA, USA June. 2025 – Nov. 2025

- Topic: high compression ratio image tokenizer for generation.
- Worked with Zhoutong Zhang.

Research Scientist Intern **Adobe Nextcam**, San Jose, CA, USA June. 2024 – Sep. 2024

- Topic: mobile burst matting from the fusion of parallax cues.
- Worked with Marc Levoy's computational photography team.

Research Assistant **MMLab@CUHK**, Hong Kong, China Sep. 2023 – Present

- Topic: generative model for inverse imaging problems.
- Worked with Prof. Tianfan Xue.

Publications

- Kaixiong Gong*, **Xin Cai***, Bin Lin, Hao Wang, Yunlong Lin, Mingzhe Zheng, Bohao Li, Jian-Wei Zhang, Miles Yang, Zhao Zhong, Liefeng Bo, Xiangyu Yue. "Twins: Learn to Predict Unified Representations with Focal Loss." International Conference on Machine Learning (ICML), 2026.
- **Xin Cai**, Zhiyuan You, Zhoutong Zhang, Tianfan Xue. "DA-VAE: Plug-in Latent Compression for Diffusion via Detail Alignment." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- **Xin Cai**, Jiawen Chen, Lars Jebe, Tianfan Xue, Zhoutong Zhang. "Parallax Portrait Matting." European Conference on Computer Vision (ECCV), 2026.
- **Xin Cai**, Hailong Zhang, Chenchen Wang, Wentao Liu, Jinwei Gu, Tianfan Xue. "LenslessFace: An End-to-End Optimized Lensless System for Privacy-Preserving Face Verification." IEEE Transactions on Image Processing (TIP), 2026.
- **Xin Cai**, Zhiyuan You, Hailong Zhang, Wentao Liu, Jinwei Gu, Tianfan Xue. "PhoCoLens: Photorealistic and Consistent Reconstruction in Lensless Imaging." NeurIPS, 2024, **Spotlight**.

- **Xin Cai**, Jiabei Zeng, Shiguang Shan, Xilin Chen. "Source-free Adaptive Gaze Estimation with Uncertainty Reduction." The IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR), 2023.
- **Xin Cai**, Jiabei Zeng, Shiguang Shan. "Landmark-aware Self-supervised Eye Semantic Segmentation." IEEE International Conference on Automatic Face and Gesture Recognition (FG), 2021.
- **Xin Cai**, Boyu Chen, Jiabei Zeng, Jiajun Zhang, Yunjia Sun, Xiao Wang, Zhilong Ji, Xiao Liu, Xilin Chen, Shiguang Shan. "Gaze Estimation with an Ensemble of Four Architectures." *arXiv:2107.01980*, 2021. (Technical report for the **winner** solution in ETH-XGaze Gaze Estimation Challenge@CVPR 2021)
- Yichen Bian, Zixuan Chen, Yujin Wang, **Xin Cai**, Shi Guo, Junhao Zhuang, Leilei Gu, Tianfan Xue. "Swift-Fusion: Towards Motion-Robust and Efficient 4K HDR Fusion." ACM SIGGRAPH Asia (TOG), 2026.
- Yutian Chen, Shi Guo, Renbiao Jin, Tianshuo Yang, **Xin Cai**, Yawen Luo, Mingxin Yang, Mulin Yu, Linning Xu, Tianfan Xue. "AnyRecon: Arbitrary-View 3D Reconstruction with Video Diffusion Model." ACM SIGGRAPH Asia, 2026.
- Junhao Zhuang, Shi Guo, **Xin Cai**, Xiaohui Li, Yihao Liu, Chun Yuan, Tianfan Xue. "FlashVSR: Towards Real-Time Diffusion-Based Streaming Video Super-Resolution." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- Zhiyuan You, Ke Wang, He Zhang, **Xin Cai**, Jinjin Gu, Tianfan Xue, Chao Dong, Zhoutong Zhang. "PhotoFramer: Multi-modal Image Composition Instruction." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- Chenjian Gao, Lihe Ding, **Xin Cai**, Zhanpeng Huang, Zibin Wang, Tianfan Xue. "LoRA-Edit: Controllable First-Frame-Guided Video Editing via Mask-Aware LoRA Fine-Tuning." International Conference on Learning Representations (ICLR), 2026.
- Zhiyuan You, **Xin Cai**, Jinjin Gu, Tianfan Xue, Chao Dong. "Teaching Large Language Models to Regress Accurate Image Quality Scores using Score Distribution." The IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025.
- Zixuan Chen, Yujin Wang, **Xin Cai**, Zhiyuan You, Zheming Lu, Fan Zhang, Shi Guo, Tianfan Xue. "UltraFusion: Ultra High Dynamic Imaging using Exposure Fusion." The IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR), 2025, **Highlight, Best Demo Honorable Mention**.
- Zhiyuan You, Jinjin Gu, **Xin Cai**, Zheyuan Li, Kaiwen Zhu, Chao Dong, Tianfan Xue. "Descriptive Image Quality Assessment in the Wild." IEEE Transactions on Image Processing (TIP), 2025.
- Lifan Xia, Yong Li, **Xin Cai**, Zhen Cui, Chunyan Xu, Antoni B Chan. "Collaborative Contrastive Learning for Cross-domain Gaze Estimation." Pattern Recognition 161:111244, 2025.
- Ken Deng, Yuanchen Guo, Jingxiang Sun, Zixin Zou, Yangguang Li, **Xin Cai**, Yanpei Cao, Yebin Liu, Ding Liang. "DetailGen3D: Generative 3D Geometry Enhancement via Data-Dependent Flow." 3DV, 2025.

Teaching

Teaching Assistant, CSCI2100B Data Structures — Fall 2023, CUHK
Teaching Assistant, IERG2080/ESTR2306 – Introduction to Systems Programming — Spring 2024, CUHK
Teaching Assistant, IERG5670 – Computational Imaging Systems and Algorithms — Fall 2024, CUHK

Services

Conference Reviewer: CVPR, NeurIPS, ICLR, AAAI, ICML, ICCV
Journal Reviewer: IEEE TPAMI, IEEE TIP, IEEE TCI, CVIU

Honors and Awards

Academy Scholarship of University of Chinese Academy of Science — 2017, 2018 and 2019
The Tang Lixin Academic Excellence Scholarship — Jun. 2020
1st Prize, Award on ETH-XGaze Challenge — Jun. 2021
Merit Student of University of Chinese Academy of Sciences — 2018, 2021 and 2023
Postgraduate Studentship — Mar. 2023
Outstanding Tutors Award — 2024